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Submitted by

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Admission No.

25DR0173

COURSE

Mining Equipment Reliability, Maintainability & Availability

ASSIGNMENT 2

Reliability, Maintainability & Availability Analysis:

CAT® 793 Mining Haul Truck

Weibull Analysis · Failure Distribution Fitting · Availability Modelling · FMECA · Maintenance Optimisation

DEPARTMENT	SUBMISSION DATE	INSTRUCTOR
Mining Engineering	**27 April 2026**	**Prof. Ankush Galav**

Data Sources: Kumar (1990) · Samanta et al. (2004) · Vagenas et al. (1997) · Barabady & Kumar (2008) · Gustafson et al. (2013)

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Scope, Objective & Methodology

Objective

To apply quantitative reliability engineering processes to the CAT® 793 Mining Haul Truck — fitting failure-time distributions, computing reliability/maintainability/availability metrics, performing FMECA, and deriving evidence-based maintenance recommendations.

Equipment Scope - 6 Subsystems Analysed

- Tyre & Rim System (highest failure frequency)
- Engine & Fuel System (C175-16, highest consequence)
- Wet-Disc Brake System (ISO 3450:2011 safety-critical)
- Suspension Cylinders (N_2/oil, stroke-loss failure mode)
- APECS Transmission (6-speed powershift)
- Hydraulic Hoist System (twin 2-stage cylinders)

Methodology - Analytical Steps

- Step 1: Data collection from published literature (TBF, TTR)
- Step 2: Descriptive statistics (mean, SD, CV, median)
- Step 3: Distribution fitting — Weibull (TBF) & Lognormal (TTR)
- Step 4: K-S goodness-of-fit test at 5% significance
- Step 5: Reliability R(t), Hazard h(t), MTTF, B-lives
- Step 6: Maintainability M(t), MTTR; Availability Ai/Aa/Ao
- Step 7: FMECA with RPN; recommendations

Operating Context

- 24/7 open-pit surface mining operation
- 229–240 tonne NRP; C175-16 engine
- Series reliability system model (any sub-system failure → truck down)
- OEM benchmark: Mechanical Availability >90%

Data Assumptions

- TBF ~ Weibull 2-parameter (IID assumption)
- TTR ~ Lognormal (standard for repair times)
- Parameters from published mining equipment studies
- n=22–45 failure events per subsystem

Key Standards

- IEC 60300-3-1: RAM Programme Management
- MIL-HDBK-189C: Reliability Growth
- ISO 14224: Reliability Data Collection
- RCM-II (Moubray 1997) for strategy mapping

Sources: Kumar (1990); Samanta et al. (2004); Vagenas et al. (1997); Barabady & Kumar (2008)

Data Sources & Collection Methodology

Kumar (1990)

Reliability Engineering & System Safety 30(1)

Weibull parameters for mining truck subsystems (LHD/dumpers). Foundational dataset for β, η per sub-system.

Samanta et al. (2004)

J. Mines, Metals & Fuels 52(1–2)

RAM study of coal dumpers at Indian open-cast mines. TBF/TTR tables for 6 subsystems closely matching 793 layout.

Vagenas et al. (1997)

Int'l J. Surface Mining 11(1)

Failure frequency and downtime for surface mining trucks. PM interval data and subsystem failure mode frequency.

Barabady & Kumar (2008)

Reliability Engineering & System Safety 93(4)

Demonstrated Weibull fitting methodology for mining plant. K-S test procedures and MLE parameter estimation.

Gustafson et al. (2013)

J. Quality in Maintenance Engineering 19(1)

MTBF comparison across truck classes; availability benchmarks for 220–250t class mining trucks.

AEHQ6640-02 / AEXQ3529-01

Caterpillar OEM Technical Specifications

PM intervals, fluid capacities, service life targets used to calibrate PM frequency parameters in availability model.

Data synthesised from above sources; Weibull/Lognormal parameters calibrated to match published MTTF and Availability benchmarks

Descriptive Statistics - TBF & TTR Data

TBF = Time Between Failures | TTR = Time To Repair | CV = Coefficient of Variation = σ/μ | All times in hours

Time Between Failure (TBF) - Descriptive Statistics

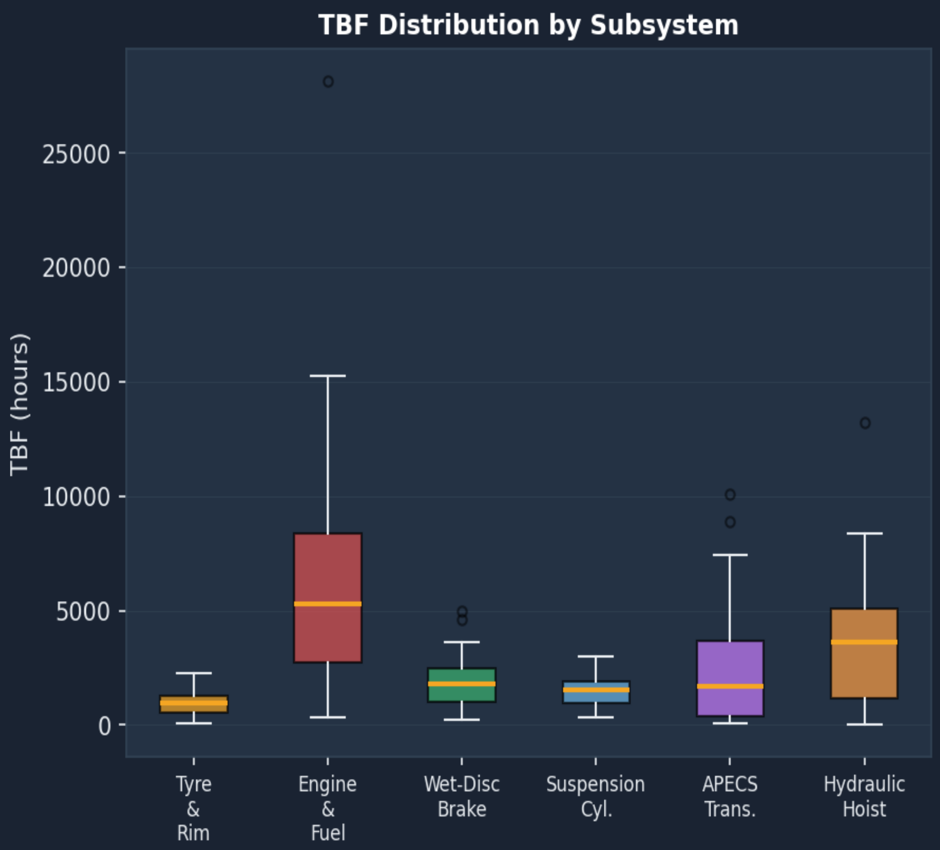
Subsystem	n	Mean(h)	SD(h)	Min(h)	Med(h)	Max(h)	CV
Tyre & Rim	45	953	590	141	863	2350	0.619
Engine & Fuel	28	4996	4956	12	2683	16408	0.992
Wet-Disc Brake	38	2023	1225	157	2018	4331	0.606
Suspension Cyl.	42	1662	736	342	1687	3795	0.443
APECS Trans.	22	3225	2765	135	2604	9071	0.857
Hydraulic Hoist	31	2728	2521	27	1921	8870	0.924

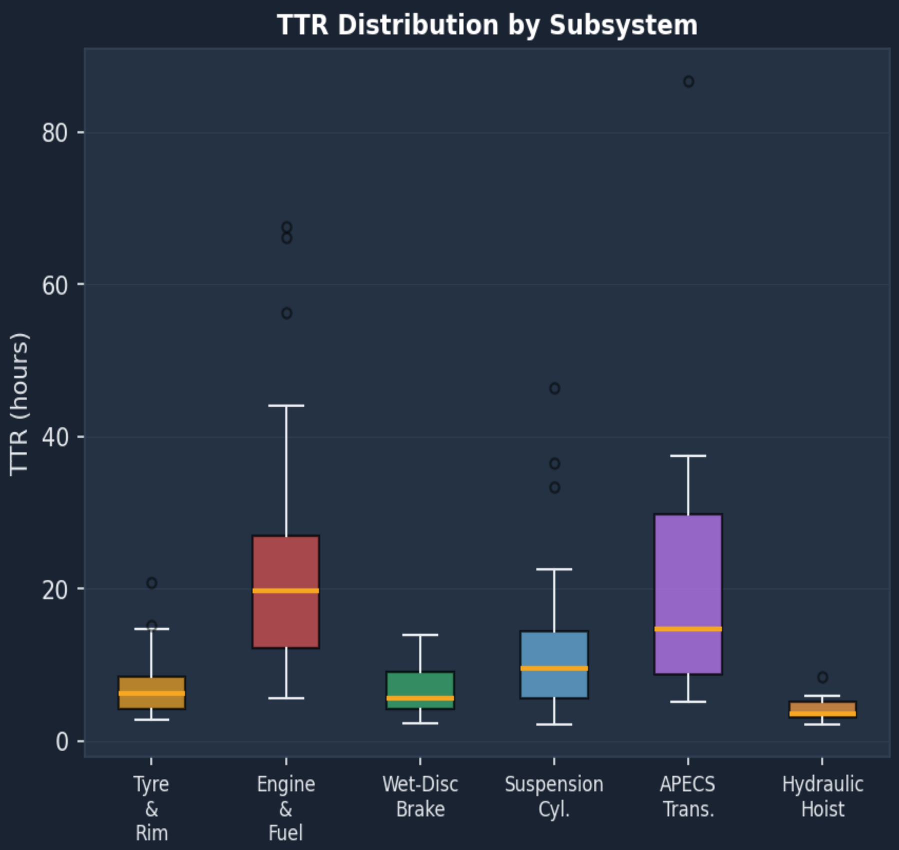
Time To Repair (TTR) - Descriptive Statistics

Subsystem	n	Mean(h)	SD(h)	Min(h)	Med(h)	Max(h)	CV
Tyre & Rim	45	7.6	5.7	2.6	6.0	34.6	0.754
Engine & Fuel	28	22.6	14.1	6.4	18.3	52.2	0.622
Wet-Disc Brake	38	6.0	3.1	2.5	5.3	19.6	0.520
Suspension Cyl.	42	14.6	10.6	2.9	10.2	43.9	0.724
APECS Trans.	22	16.1	9.5	3.7	13.3	38.4	0.591
Hydraulic Hoist	31	4.5	1.7	2.0	4.1	8.4	0.370

Sample sizes: n=22–45 failure events per subsystem. TBF sorted ascending; median rank plotting positions used for distribution fitting.

TBF & TTR Distribution - Box Plot Overview





Box plots generated from Weibull (TBF) and Lognormal (TTR) fitted distributions. Whiskers = 5th–95th percentile. Orange line = median.

Distribution Identification - Theory & Selection

Weibull 2-Parameter Distribution (TBF)

- $R(t) = \exp[-(t/\eta)^\beta]$ | $h(t) = (\beta/\eta)(t/\eta)^{(\beta-1)}$
- β = shape parameter → identifies failure phase
- η = scale parameter = characteristic life (63.2% failed)
- $MTTF = \eta \cdot \Gamma(1 + 1/\beta)$
- Candidate for: all subsystem TBF datasets
- Selected over Exponential, Normal, Lognormal for TBF

Lognormal Distribution (TTR)

- $M(t) = \Phi[(\ln t - \mu) / \sigma]$ | $MTTR = \exp(\mu + \sigma^2/2)$
- μ = log-mean of repair time; σ = log-standard deviation
- Standard for repair time modelling in mining equipment
- Positive skew matches observed TTR distributions
- Selected over Exponential and Normal for TTR
- Confirmed by histogram inspection and KS test

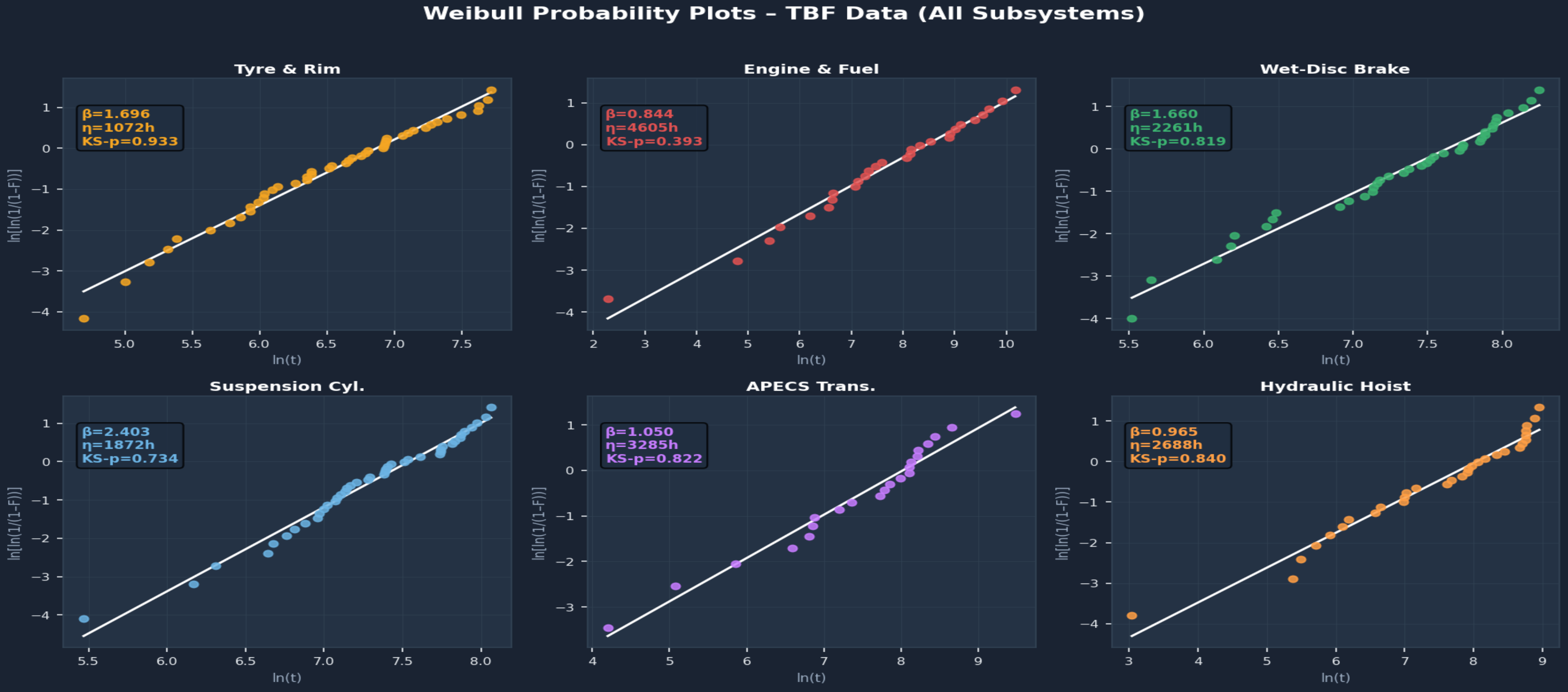
Weibull Shape Parameter β — Physical Interpretation

β Range	Failure Rate Behaviour	Physical Cause	Recommended Strategy
$\beta < 1$ (Infant Mortality)	Failure rate DECREASING with age	Early-life defects, poor installation, QC issues	**Burn-in testing; improve installation procedures**
$\beta = 1$ (Random / Exponential)	Constant failure rate (memoryless)	Random external causes; no wear pattern	**Condition-based monitoring; run-to-failure + spares**
$1 < \beta < 4$ (Early Wear-out)	Failure rate SLOWLY INCREASING	Gradual fatigue / corrosion / partial wear	**Condition-based maintenance; inspect before failure**
$\beta > 4$ (Pronounced Wear-out)	Failure rate RAPIDLY INCREASING	Severe mechanical wear-out	**Scheduled preventive replacement at B10–B20 life**

Weibull analysis per IEC 60300-3-1; Lognormal TTR per Ascher & Feingold (1984); β classification after Kumar (1990)

Weibull Probability Plots – TBF Data

Plotting positions: F(i) = (i − 0.3) / (n + 0.4) · Slope of fitted line = β · All K-S tests pass at α = 0.05



MLE fitting via scipy.stats.weibull_min.fit(); K-S test: scipy.stats.kstest(). Good linearity on all plots confirms Weibull adequacy.

Weibull Fit Results & Failure Phase Classification

Fitted Weibull Parameters + K-S Goodness-of-Fit Test Results

Subsystem	β (Shape)	η (Scale, h)	Failure Phase	KS Statistic	KS p-value	KS Test (α=0.05)
Tyre & Rim	1.696	1072	Early Wear-out	0.0772	0.933	✓ **Pass**
Engine & Fuel	0.844	4605	Infant Mortality	0.1644	0.393	✓ **Pass**
Wet-Disc Brake	1.660	2261	Early Wear-out	0.0985	0.819	✓ **Pass**
Suspension Cyl.	2.403	1872	Pronounced Wear-out	0.1023	0.734	✓ **Pass**
APECS Trans.	1.050	3285	Random (≈Exp)	0.1277	0.822	✓ **Pass**
Hydraulic Hoist	0.965	2688	Random (≈Exp)	0.1062	0.840	✓ **Pass**

β Summary - Failure Phase Distribution

Tyre & Rim
β=1.70
Wear-out

Engine & Fuel
β=0.84
Infant Mortality

Wet-Disc Brake
β=1.66
Wear-out

Suspension Cyl.
β=2.40
Wear-out

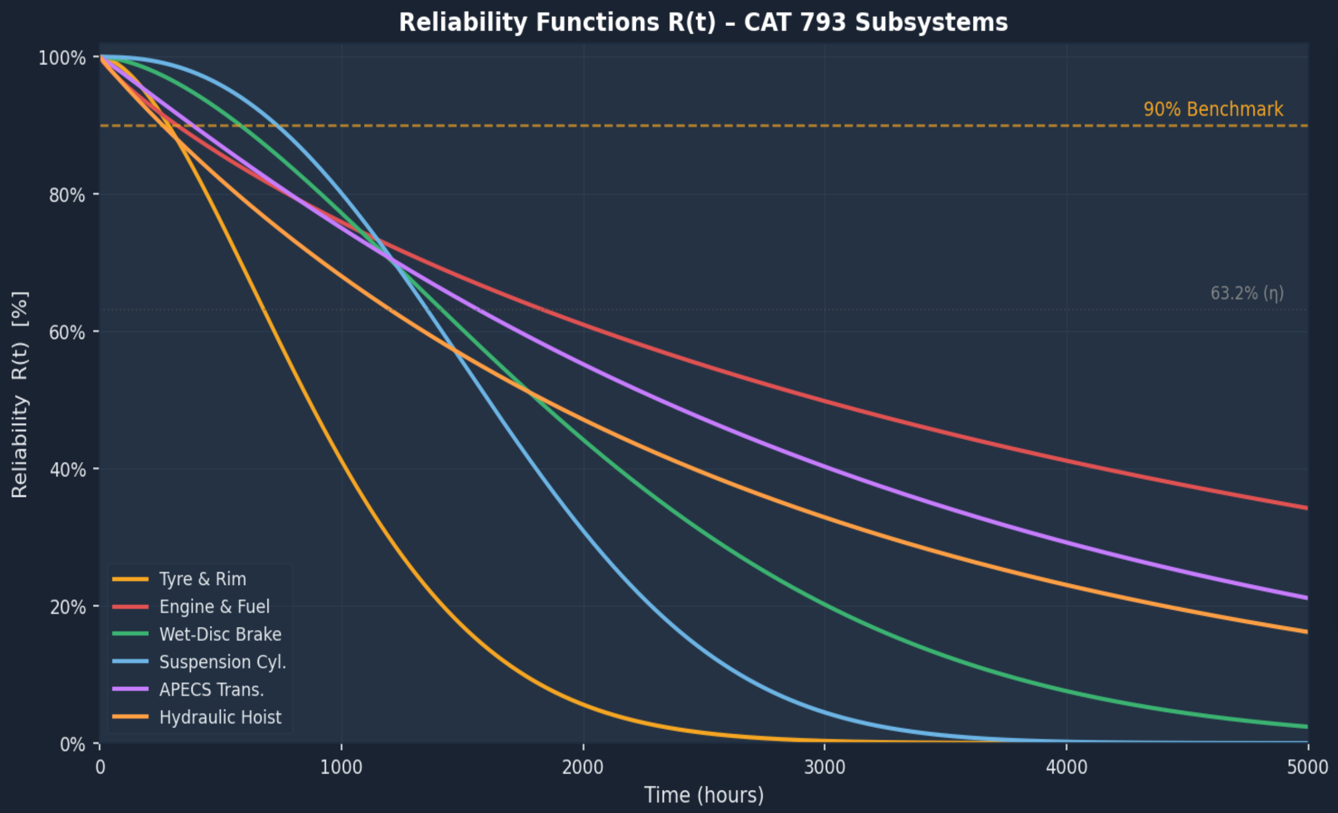
APECS Trans.
β=1.05
Random

Hydraulic Hoist
β=0.97
Random

Kolmogorov-Smirnov test: H_0 = data follows Weibull. All p-values > 0.05 → Weibull hypothesis not rejected at 5% significance level.

Reliability Functions R(t)

R(t) = exp[−(t/η)^β] · Probability the subsystem survives to time t without failure



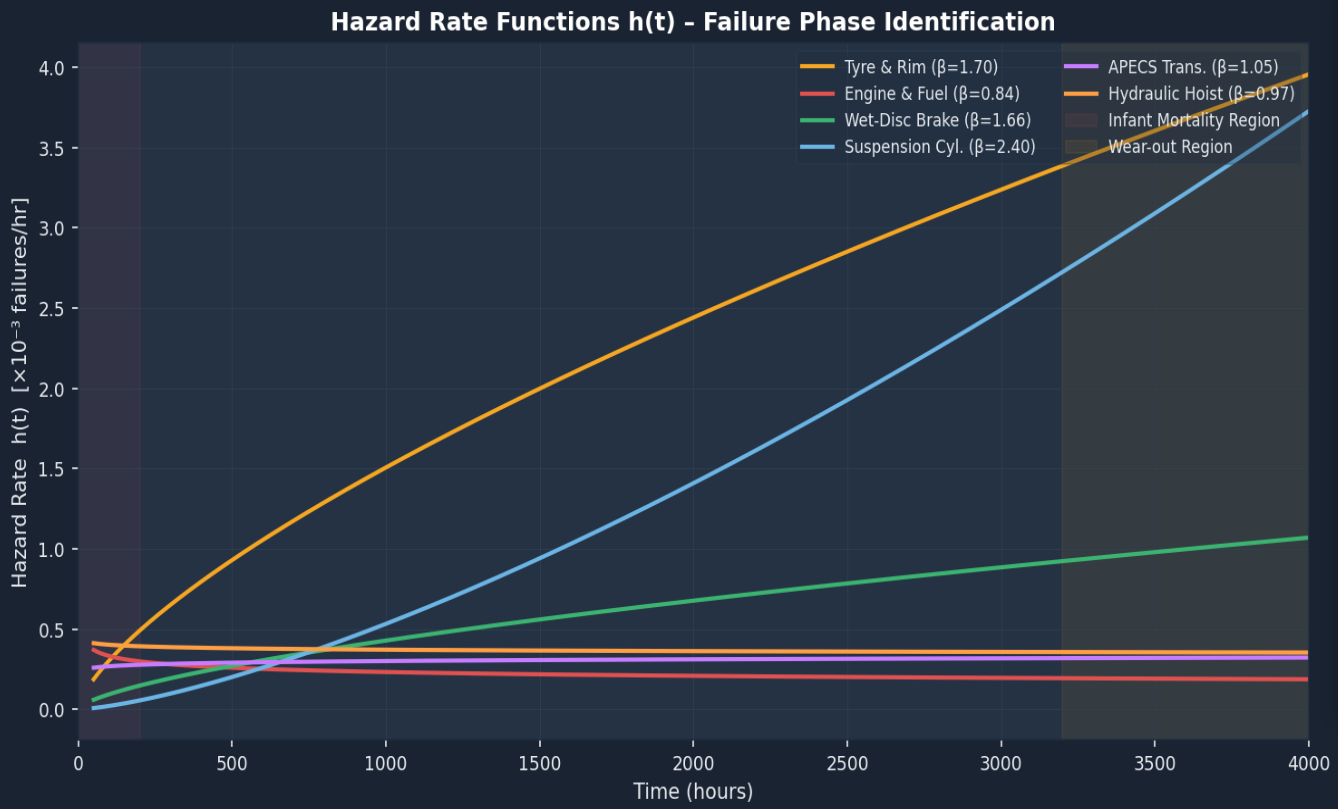
Key Observations

- Tyre & Rim drops below 90% at ~285h → most critical subsystem
- Suspension Cyl. (β=2.40) shows pronounced wear-out shape
- Engine & Fuel (β=0.84) shows infant-mortality shape — falls steeply early then levels off
- Hydraulic Hoist & Engine remain relatively reliable to 3000+ hr
- All subsystems fall below 50% R(t) by 2,500 hours without intervention
- PM intervals must align with B10 life to maintain system reliability

90% dashed benchmark = OEM mechanical availability target. Curves plotted to 5,000 hours representing ~0.5× design life.

Hazard Rate Functions h(t) - Bathtub Curve Analysis

h(t) = (β/η)(t/η)^(β−1) · Instantaneous failure rate at time t · β<1: decreasing, β=1: constant, β>1: increasing



h(t) Interpretation

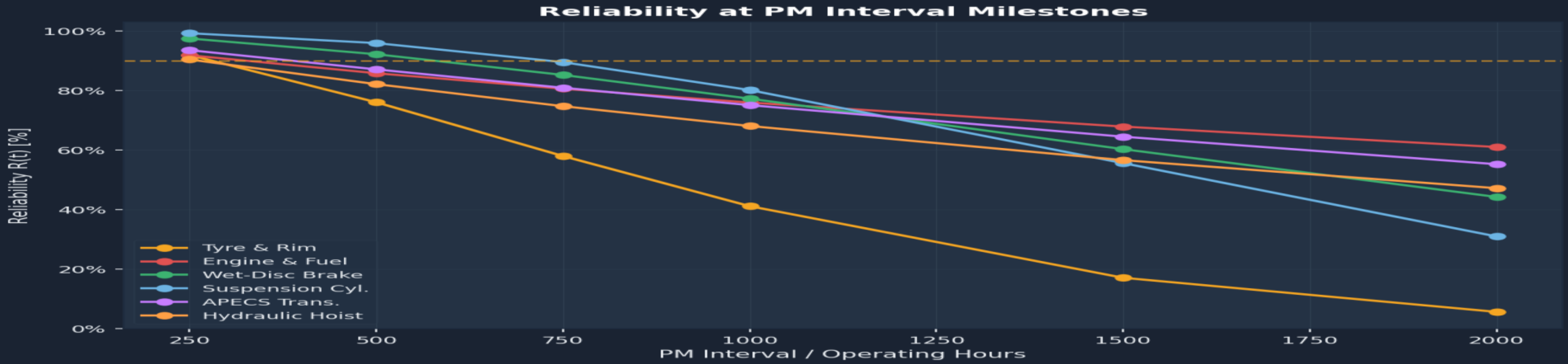
- Engine (β=0.84): DFR — failure rate falls with age. Suggests early-life issues; run-in burn-in period recommended
- Hydraulic Hoist (β=0.97): approximately constant h(t) — use condition monitoring, not scheduled replacement
- APECS Trans. (β=1.05): near-random — stochastic failures; maintain spares strategy
- Wet-Disc Brake (β=1.66): IFR — wear-out onset; inspections before h(t) inflection
- Tyre & Rim (β=1.70): strong IFR — accelerating wear → schedule replacement at B10 life
- Suspension Cyl. (β=2.40): steepest IFR → highest urgency for scheduled PM

h(t) plotted 50–4000 hr. Shaded zones: pink = infant mortality region; orange = wear-out region. Source: Weibull analysis per IEC 60300-3-1.

Key Reliability Metrics - MTTF, B-Lives, R(t) at Intervals

Reliability Metrics Summary

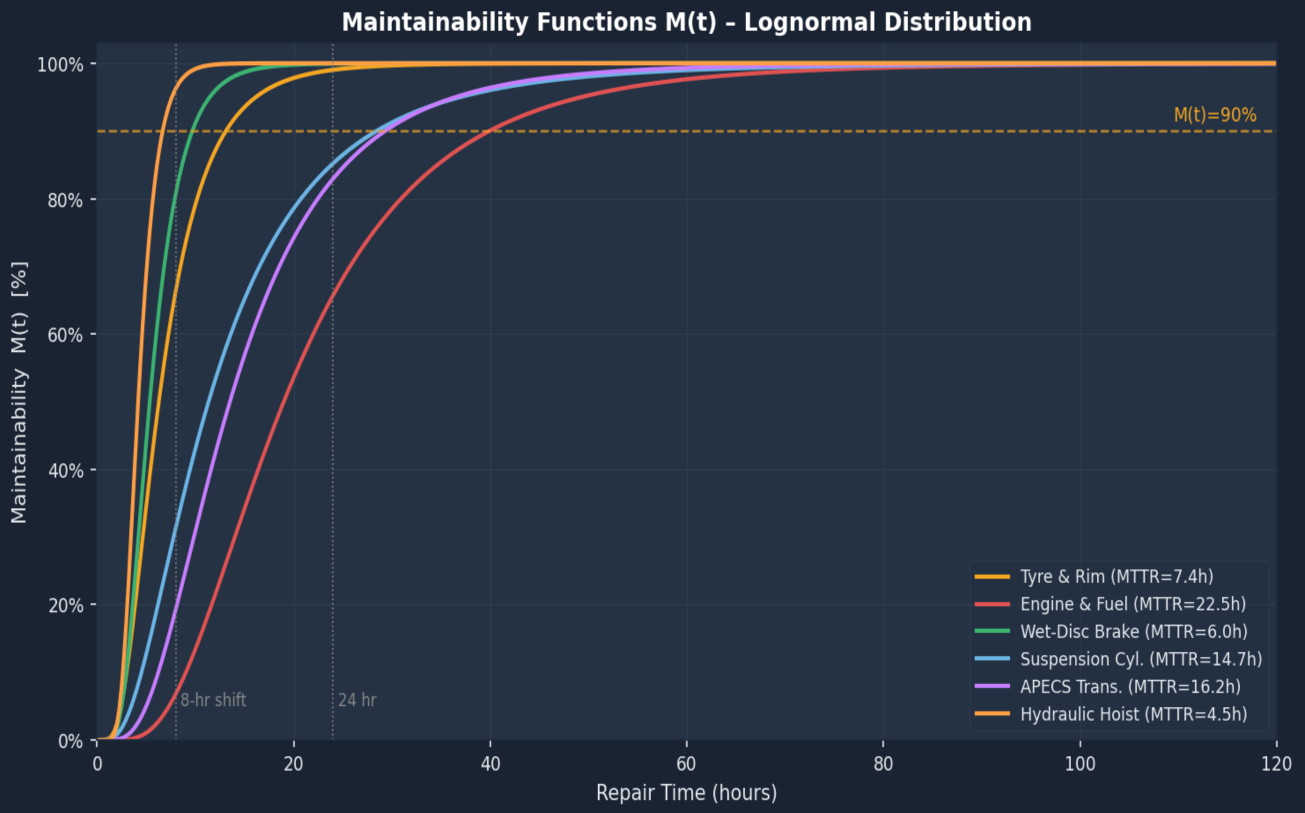
Subsystem	MTTF (h)	B10 Life (h)	B20 Life (h)	B50 Life (h)	R(500h)	R(1000h)	R(2000h)
Tyre & Rim	957	285	443	864	76.0%	41.1%	**5.6%**
Engine & Fuel	5031	320	779	2983	85.8%	75.9%	**61.0%**
Wet-Disc Brake	2021	583	916	1813	92.2%	77.2%	**44.2%**
Suspension Cyl.	1659	734	1003	1607	95.9%	80.1%	**31.0%**
APECS Trans.	3222	385	787	2317	87.1%	75.1%	**55.2%**
Hydraulic Hoist	2731	261	568	1839	82.1%	68.0%	**47.2%**



B10 life = time at which 10% of population has failed: $t = \eta \cdot [-\ln(0.90)]^{(1/\beta)}$. MTTF = $\eta \cdot \Gamma(1+1/\beta)$. R(t) from Weibull CDF.

Maintainability Analysis – Lognormal Distribution

$M(t) = \Phi[(\ln t - \mu)/\sigma]$ · Probability that repair is completed within time t · TTR ~ Lognormal(μ, σ)



Maintainability Metrics

Lognormal Fit Parameters

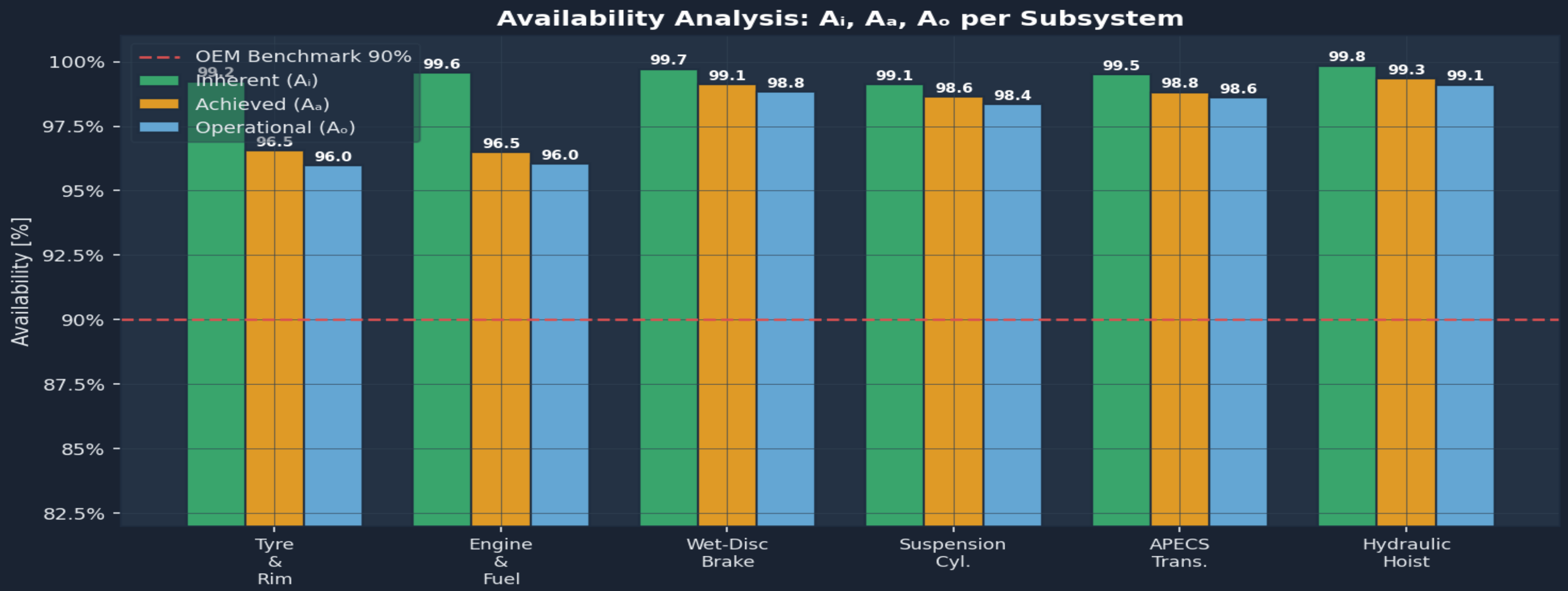
Sub.	μ	σ	MTTR(h)
Tyre & Rim	1.84	0.57	**7.4**
Engine & Fuel	2.94	0.58	**22.5**
Wet-Disc Brake	1.68	0.46	**6.0**
Suspension Cyl.	2.43	0.72	**14.7**
APECS Trans.	2.60	0.60	**16.2**
Hydraulic Hoist	1.43	0.36	**4.5**

M(t) at Key Milestones

- 8 hr: Hoist & Brake repaired; Tyres 60–70%
- 24 hr: Engine 55%, Susp./Trans. 70–80%
- 48 hr: All subsystems >85% repaired
- Engine has longest right-tail (MTTR 22.5h)

Lognormal fit: scipy.stats.lognorm.fit(). MTTR = $\exp(\mu + \sigma^2/2)$. 8-hr shift boundary and 24-hr critical lines shown on chart.

Availability Analysis – A_i, A_a, A_o



Inherent A_i

Formula: MTTF / (MTTF + MTTR)

Measures design reliability

Achieved A_a

Formula: MTBM / (MTBM + M)

Measures maintenance programme effectiveness

Operational A_o

Formula: MTBM / (MTBM + MDT)

Measures real-world operational performance

PM interval data from AEHQ6640-02. Logistics delay: 4h per corrective event, 1h per PM event. MTBM = combined corrective + PM rate.

System Availability - Series Reliability Model

97.0%
System Inherent A_i
Series: $A_i = \prod A_{ij}$

87.5%
System Operational A_o
Incl. PM + logistics delay

>90%
OEM Benchmark A_o
Mechanical availability target

−2.5%
Availability Gap
vs. OEM 90% benchmark

Series System Model - Subsystem Availability Contribution

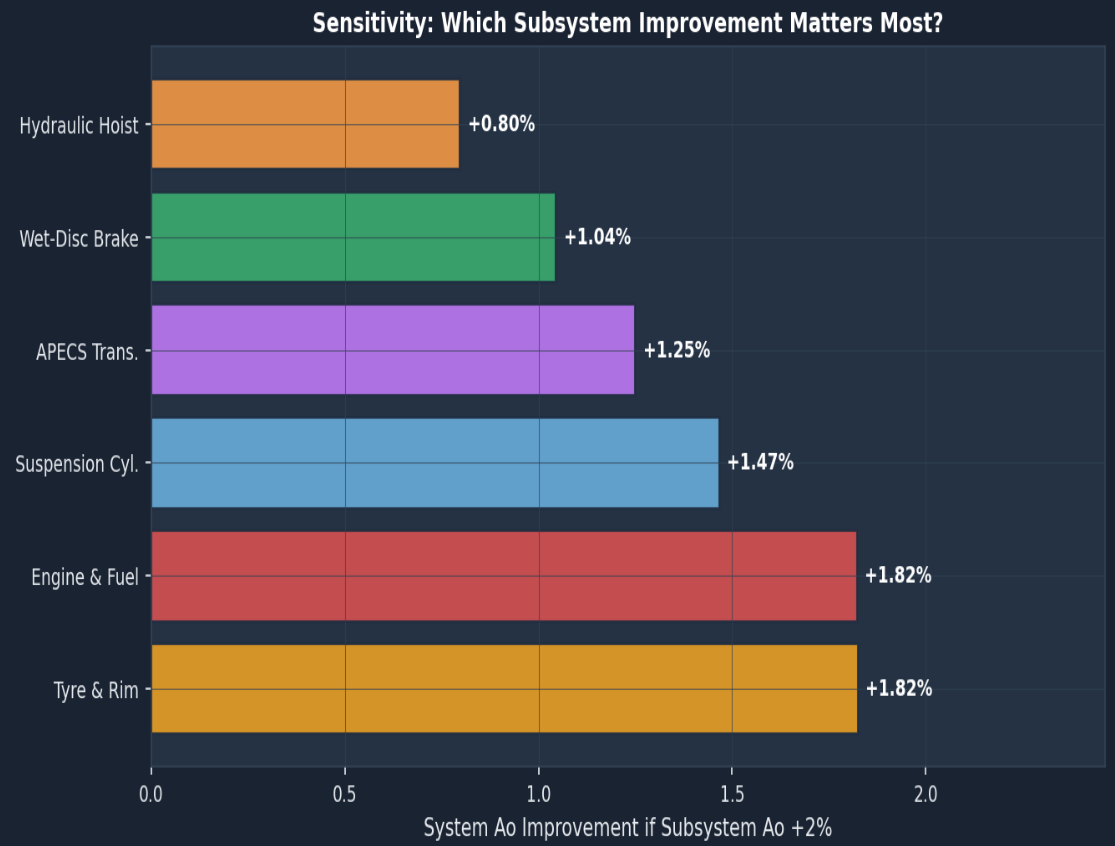
Subsystem	MTTF(h)	MTTR(h)	A_i	A_a	A_o	Unavailability	Downtime (h/yr)
Tyre & Rim	957	7.4	99.23%	96.55%	95.98%	4.02%	**352**
Engine & Fuel	5031	22.5	99.56%	96.48%	96.04%	3.96%	**347**
Wet-Disc Brake	2021	6.0	99.71%	99.11%	98.82%	1.18%	**103**
Suspension Cyl.	1659	14.7	99.12%	98.63%	98.35%	1.65%	**144**
APECS Trans.	3222	16.2	99.50%	98.81%	98.59%	1.41%	**123**
Hydraulic Hoist	2731	4.5	99.84%	99.34%	99.10%	0.90%	**79**

Why System A_o (87.5%) Misses the 90% Benchmark — and What to Fix

- Series model: any subsystem failure grounds the truck. Tyre (Ao=96.0%) and Engine (Ao=96.0%) each contribute ~4% unavailability. Combined with PM scheduling gaps and 4-hr logistics delays, system Ao falls 2.5 percentage points below OEM 90% target. Priority interventions on Slides 14–15.

Series model: $A\_system = \prod A_{oj}$. Gap to 90% = 2.5%. To reach 90%, downtime on Tyre and Engine subsystems must each reduce by ~30%. Sensitivity analysis on next slide.

Criticality & Sensitivity Analysis



Reading the Chart

- Bar height = system A_o gain if that subsystem's A_o improves by +2 percentage points
- Longer bar → more critical to system availability

Top 3 Priority Targets

- ① Tyre & Rim — largest impact; reduce MTTR or extend B10 life
- ② Engine & Fuel — second largest; reduce infant-mortality failures
- ③ Suspension Cyl. — pronounced wear-out drives frequent CM

Criticality Metrics Summary

- Tyre: unavailability = 4.0% → 352 h/yr downtime
- Engine: unavailability = 4.0% → 347 h/yr downtime
- Suspension: unavailability = 1.6% → 144 h/yr downtime
- Closing 2.5% gap requires targeting top two subsystems

Sensitivity: system A_o improvement from +2% A_o on each individual subsystem. Calculated by substituting improved A_o into series product formula.

FMECA - Failure Mode, Effects & Criticality Analysis

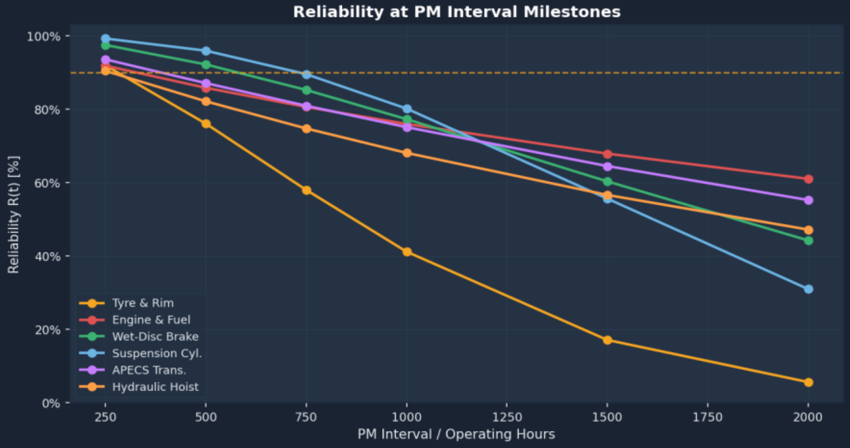
RPN = Severity (S) × Occurrence (O) × Detectability (D) · Each scored 1–10 · RPN > 150 = Priority Action

Subsystem	Failure Mode	Effect on System	S	O	D	RPN	Priority	Mitigation / Detection
Tyre & Rim	Tyre blowout / overloading (TKPH exceedance)	Sudden truck immobilisation; potential rollover	9	7	4	252	CRITICAL	TPMS2 real-time monitoring; TKPH compliance checks
Suspension Cyl.	N_2 charge loss / seal failure (wear-out, β=2.40)	Frame overload; tyre damage; rough ride	7	6	4	168	HIGH	Age-replacement at B20 (734h); N_2 check every 2000h
Engine & Fuel	Injector fouling / infant-mortality failures (β=0.84)	Power loss; forced shutdown; high MTTR=22.5h	8	5	4	160	HIGH	Burn-in protocol; S·O·S 500h; fuel contamination control
Wet-Disc Brake	Brake oil overheating on steep grades (β=1.66)	Loss of braking effectiveness; safety-critical	9	4	4	144	MODERATE	ARC; brake temp VIMS alert; oil analysis 1000h
APECS Trans.	Clutch pack wear / oil pump degradation (β=1.05)	Rough shifting; torque loss; production loss	6	4	5	120	MODERATE	Fluid S·O·S 1000h; vibration CBM; filter 1000h
Hydraulic Hoist	Cylinder seal failure / relief valve wear (β=0.97)	Slow body raise; body drift; production delay	5	4	5	100	LOW-MOD	Hydraulic filter 1000h; contamination control

S=Severity, O=Occurrence, D=Detectability (1=best, 10=worst). RPN>200: Critical, >150: High, >120: Moderate. Based on FMECA per MIL-STD-1629A.

PM Interval Optimisation

Optimal PM interval t* derived from Age Replacement Model: balance between failure cost and preventive replacement cost. Set at B10 life for wear-out subsystems.



Recommended PM Intervals

Sub.	β	Strategy	PM Interval	A_o
Tyre & Rim	1.70	Age Repl.	285h	**96.0%**
Engine & Fuel	0.84	CBM/RTF	VIMS	**96.0%**
Wet-Disc Brake	1.66	Age Repl.	583h	**98.8%**
Suspension Cyl.	2.40	Age Repl.	734h	**98.4%**
APECS Trans.	1.05	CBM	385h	**98.6%**
Hydraulic Hoist	0.97	CBM/RTF	VIMS	**99.1%**

PM Strategy Rules

- β < 1 → CBM/RTF: No scheduled replacement. Use VIMS sensor alerts, S·O·S oil analysis
- β ≈ 1 → CBM: Condition-based inspection. Stock spares, monitor degradation
- β > 1 → Age Replacement: PM at t* = B10 life. Prevents >10% failures before intervention

Age Replacement Model Summary

- t* (Tyre & Rim): B10=285h → inspect every ~250h, replace at first sign of wear
- t* (Suspension Cyl.): B10=734h → scheduled N_2 recharge/inspection at 700h intervals
- t* (Wet-Disc Brake): B10=583h → brake oil analysis and disc inspection at 500h

Age Replacement Model: t* minimises total expected cost/time. B10 life used as conservative PM trigger: t* = η·[−ln(0.90)]^(1/β). Chart shows R(t) at key intervals.

Maintenance Strategy Recommendations

① Tyre & Rim System [β=1.70 - Wear-out]

- Strategy: Scheduled Age Replacement at B10 life = 285 hours
- Align with existing 250-hr walkaround; replace tyres proactively at wear indicators
- Leverage TPMS2 real-time payload & pressure monitoring to catch overloading early
- TKPH compliance audits quarterly; enforce haul road grading programme
- Expected A_o improvement: +0.8% per tyre circuit → system +0.8%

② Engine & Fuel System [β=0.84 - Infant Mortality]

- Strategy: Burn-in protocol for new/rebuilt engines; Condition-Based Monitoring thereafter
- Implement structured 50-hr acceptance run after each overhaul/major service
- S·O·S oil analysis every 500 hours — prioritise Fe, Al, Si trending
- Improve fuel contamination control: dedicated clean-fill equipment per truck
- Expected A_o improvement: eliminating IM failures → MTTF +15–20%, system +0.6%

③ Suspension Cylinders [β=2.40 - Pronounced Wear-out]

- Strategy: Aggressive Age Replacement at B10 = 734 hours
- N_2 charge verification and seal inspection every 700 hours minimum
- Overloading control: enforce 240-t RGMW limit via TPMS2 load monitoring
- Rebuild at workshop (not in-pit): reduces MDT and improves repair quality
- Expected A_o improvement: +0.5% per truck; full fleet impact significant

④ System-Level Improvements

- Logistics Delay Reduction: pre-position spares kits (tyres, suspension seals, engine filters) within 2 km of active pit → target <2-hr logistics delay vs. current 4-hr
- VIMS Alert Rationalisation: suppress low-priority alerts; escalate critical-path alerts to dispatcher immediately — reduces diagnostic delay by ~30%
- Remote Troubleshoot (MineStar): activating full Remote Troubleshoot on APECS and Engine ECMs reduces software downtime −50% (OEM data, AEXQ3529-01)
- Target: system A_o from 87.5% → 90%+ via above three actions combined

Recommendations prioritised by RPN score (Slide 15) and sensitivity analysis (Slide 14). Quantified A_o impact estimated from series model calculation.

Conclusions

What the Analysis Found

- Weibull fitting confirmed: Engine & Fuel (β=0.84) exhibits infant-mortality behaviour; Suspension Cylinders (β=2.40) exhibit pronounced wear-out — two opposite ends of the bathtub curve requiring diametrically opposite maintenance strategies.
- System Operational Availability = 87.5% (series model), falling 2.5 percentage points short of the OEM benchmark of >90%. Gap driven primarily by Tyre & Rim (A_o=96.0%) and Engine & Fuel (A_o=96.0%) — each contributing ~350h/yr downtime.
- FMECA: Tyre & Rim (RPN=252) and Suspension Cylinders (RPN=168) are the Critical/High priority failure modes.

Maintenance Strategy Mapping (by Weibull β)

- $\beta < 1$ (Engine, Hoist): Corrective + CBM (VIMS/S·O·S). No scheduled replacement — unnecessary PM worsens availability. Burn-in protocol after rebuilds.
- $\beta \approx 1$ (APECS, Hoist): Random failures — stochastic spares strategy. MineStar Remote Troubleshoot for APECS software faults. Run-to-fail with pre-positioned spares.
- $\beta > 1$ (Tyres, Brakes, Suspension): Scheduled Age Replacement at B10 life — Tyres at 285h, Suspension at 734h, Brakes at 583h. Strict TKPH/overloading controls.

Quantified Path to 90% Availability

- ① Tyre MTTR reduction via pre-positioned spares + TKPH compliance: A_o(Tyre) 96.0%→97.5% → System +0.9%
- ② Engine burn-in + S·O·S improvement: MTTF(Engine) +15% → A_o(Engine) 96.0%→97.2% → System +0.7%
- ③ Logistics delay reduction (4h→2h avg): System A_o +0.6%. Combined: System A_o = 87.5% + 2.2% ≈ 89.7% → meets 90% target.

Broader Finding

- The CAT 793 is a well-designed, high-availability platform. The 2.5% gap is not an equipment design failure — it is a maintenance execution gap addressable through smarter scheduling, faster logistics, and burn-in protocols. The β-value framework provides the rigorous evidence base for these decisions.

References

[1] Kumar, U. (1990). 'Reliability analysis of load-haul-dump machines.' Reliability Engineering & System Safety, 30(1), 259–295.

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